Week 8
Lecture 9 (Chap 8)
DATA GATHERING

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Aims

- Discuss how to plan and run a successful data gathering program.
- Enables you to plan and run an interview.
- Empowers you to design a simple questionnaire.
- Enables you to plan and carry out an observation.

Five key issues

1. Setting goals
    - Decide **how to** analyze data once collected
2. Identifying participants
    - Decide from **whom** to gather data
    - **How many** participants are needed
3. Relationship with participants
    - Clear and professional
    - **Informed consent** when appropriate

Five key issues

4. Triangulation

- Look at data from more than one perspective
- Collect more than one type of data, for instance, qualitative data from experiments and qualitative data from interviews
- The Power of Triangulation in Design Research | by Fjord | Design Voices | Medium

5. Pilot studies

- Small trial of main study

Capturing data

- Notes, audio, video, and photographs can be used individually or in combination:
  - Notes plus photographs
  - Audio plus photographs
  - Video
- Different challenges and advantages with each type of data recording

Sample consent form

INFORMED CONSENT FOR the Recycle Project

Please highlight your choice by clicking inside the appropriate box:

I have read and understood the information sheet, or it has been read for me, and I have been able to ask questions about my participation and my questions have been answered to my satisfaction.	YES ☐	NO ☐
I consent to be a participant in this study and understand that participation is voluntary and that I will not be paid for my participation.	YES ☐	NO ☐
I understand that I can refuse to answer questions I am not comfortable with and that I may withdraw and discontinue participation at any time in this study, up to 2 months after data collection, without giving a reason.	YES ☐	NO ☐
I understand that taking part in the study involves being observed while performing given tasks using the RECYCLE app on a smartphone provided to me, and interviewed about its use both individually and in a group.	YES ☐	NO ☐
I agree to photos being taken during the observation sessions.	YES ☐	NO ☐
I agree to the interview/focus group being audio-recorded and/or written notes being taken.	YES ☐	NO ☐
I agree to my activity on the RECYCLE app being recorded and stored in a log file.	YES ☐	NO ☐
I understand that information I provide will be used for research and dissemination purposes only.	YES ☐	NO ☐
I understand that personal information collected about me that can identify me, such as my name or where I live, will not be shared beyond the study team.	YES ☐	NO ☐
I understand that my data will be stored on encrypted devices until the end of the project when it will be destroyed.	YES ☐	NO ☐
I understand that my participation will be anonymous and any details that might identify me will not be included in reports or other publications produced from the study.	YES ☐	NO ☐
I consent for anonymized quotations from my interview to be used in reports or other publications and presentations.	YES ☐	NO ☐

Name (PRINT): Date:

Signature:

Interviews

Interviews (1)

Unstructured: Not directed by a script. Rich but not replicable.

Structured: Tightly scripted, often like a questionnaire. Replicable but may lack richness.

- "Which of the following apps do you use most frequently: Prime Video, GoogleTV, or Netflix?"
- "How often do you watch streamed content: every day, once a week, once a month, less often than once a month?"
- "Do you ever purchase anything online: Yes/No? If your answer is Yes, approximately how often do you purchase things online: every day, once a week, once a month, less frequently than once a month?"

Interviews (2)

Semi-structured: Guided by a script, but interesting issues can be explored in more depth. Can provide a good balance between richness and replicability.

Which music websites do you visit most frequently?

Answer: Mentions several but stresses that they prefer hottestmusic.com

Why?

Answer: Says that they like the site layout

Tell me more about the site layout.

Answer: Silence, followed by an answer describing the site's layout

Anything else that you like about the site?

Answer: Describes the animations

Thanks. Are there any other reasons for visiting this site so often that you haven't mentioned?

Interviews (3)

Focus groups: A group interview

- Normally, three to ten people are involved, and the discussion is led by a trained facilitator
- Participants are selected to provide a representative sample of the target population

Interview questions (1)

- Two types:
  - ‘Closed questions’ have a predetermined answer format, for example, ‘yes’ or ‘no’
  - ‘Open questions’ do not have a predetermined format

Interview questions (2)

1. Have you used a device for reading books online before? (Explore previous knowledge.)

   Interviewer checks box: ☐ *Yes* ☐ *No* ☐ *Don't remember/know*

2. Would you like to read a book using a device designed for reading online? (Explore initial reaction; then explore the response.)

   Interviewer checks box: ☐ *Yes* ☐ *No* ☐ *Don't know*

3. Why?

   If response is "Yes" or "No," interviewer asks, "Which of the following statements represents your feelings best?"

   For "Yes," interviewer checks one of these boxes:

   ☐ *I don't like carrying heavy books.*

   ☐ *This is fun/cool.*

   ☐ *My friend told me they are great.*

   ☐ *It's the way of the future.*

   ☐ *Another reason (interviewer notes the reason).*

Interview questions (3)

- What do you like most about the device? Why?
- What do you like least about the device? Why?
- Please give me an example of where the device was uncomfortable or difficult to use.

Interview questions (4)

- Avoid:
  - Long questions
  - Compound sentences — split them into two
  - Jargon and language that the interviewee may not understand
  - Leading questions that make assumptions, for example, why do you like …?
  - Unconscious biases, for instance, gender stereotypes

https://www.nngroup.com/videos/interview-mistakes-to-avoid/

Running the interview

Introduction:

Introduce yourself, explain the goals of the interview, reassure about the ethical issues, ask to record, and present the informed consent form.

Warm-up:

Make first questions easy and non-threatening.

Main body:

Present questions in a logical order

A cool-off period:

Include a few easy questions to defuse tension at the end

Closure:

Thank interviewee, signal the end, for example, switch recorder off.

Other forms of interviews

Digital conferencing systems such as Skype, Zoom, email, and smartphones can be used to conduct interviews. Some advantages are:

- Participants are in their own environment so are more relaxed
- Participants don't need to travel
- Participants don't need to worry about what to wear
- For interviews involving sensitive issues, it is easier for interviewees to be anonymous

Enriching the interview process

Props: Devices for prompting interviewee, for example, use a prototype, scenario

Questionnaires

Questionnaires

- Questions can be closed or open
- Closed questions are easier to analyze, and may be distributed and analyzed by computer
- They can be administered to large populations
- Disseminated by paper, email and the web
- Sampling can be a problem when the size of a population is unknown as is common online evaluation

Questionnaire design

- The impact of a question can be influenced by question order
- You may need different versions of the questionnaire for different populations
- Provide clear instructions on how to complete the questionnaire
- Avoid very long questions and questionnaires
- Decide on whether phrases will all be positive, all negative, or mixed

Question and response format

- 'Yes' and 'No' checkboxes
- Checkboxes that offer many options
- Rating scales
- Likert scales
- Semantic scales
- 3, 5, 7 or more points
- Use open-ended questions at the end

Question and response format: Example

What is your age?

- (●) 20 and under
- ( ) 21–30
- ( ) 31–40
- ( ) 41–50
- ( ) 51 and over

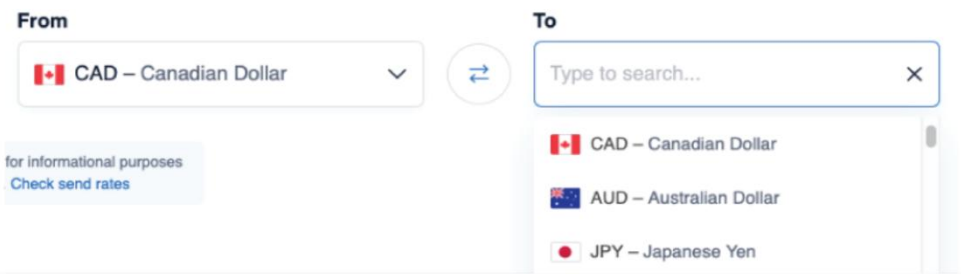
DATA GATHERING

(a)

Which news channels do you subscribe to?

- [ ] Sky News
- [x] BBC News
- [x] Al Jazeera
- [ ] Euro news
- [ ] CNN news
- [ ] Other. Please specify:_____

(b)



Question and response format: Different Scales

Strongly agree	Agree	OK	Disagree	Strongly disagree
☐	☐	☐	☐	☐

Attractive	___ ___ ___ ___ ___ ___ ___	Ugly
Clear	___ ___ ___ ___ ___ ___ ___	Confusing
Dull	___ ___ ___ ___ ___ ___ ___	Colorful
Exciting	___ ___ ___ ___ ___ ___ ___	Boring
Annoying	___ ___ ___ ___ ___ ___ ___	Pleasing
Helpful	___ ___ ___ ___ ___ ___ ___	Unhelpful
Poor	___ ___ ___ ___ ___ ___ ___	Well designed

Activity: Find 4 Problems

2. State your age in years ☐

3. How many hours a day do you spend searching online?
   - ☐ <1 hour
   - ☐ 1–3 hours
   - ☐ 3–5 hours
   - ☐ >5 hours

4. Which of the following do you do online?
   - purchase goods ☐
   - send e-mail ☐
   - visit chatrooms ☐
   - use bulletin boards ☐
   - find information ☐
   - read the news ☐

5. How useful is the Internet to you?

_____

_____

Encouraging a good response (1)

- Make sure that the purpose of study is clear
- Promise anonymity
- Ensure that questionnaire is well designed
- Offer a short version for those who do not have time to complete a long questionnaire

Encouraging a good response (2)

- If mailed, include a stamped, addressed envelope
- Follow-up with emails, phone calls, or letters
- Provide an incentive
- 40 percent response rate is good, 20 percent is often acceptable

Example of an online questionnaire

- Questionnaire shows check boxes, radio boxes, and pull-down menus

World Summit on the Information Society - Microsoft Internet Explorer

File Edit View Favorites Tools Help Back Search Favorites Folders

Address http://www.itu.int/wsis/stocktaking/scripts/q.asp Go

D. Internationally-agreed development goals outlined in the Millennium Declaration :

Is this activity relevant to achieving the MDGs listed below? (see www.un.org/millenniumgoals/ and the targets for each goal) ◉ Yes ○ No

If yes, please tick all goals that apply

1. ☐ Eradicate poverty and hunger
2. ☐ Achieve Universal Primary Education
3. ☐ Promote gender equality & empower women
4. ☑ Reduce child mortality
5. ☐ Improve maternal health
6. ☐ Combat HIV/AIDS, Malaria and other diseases
7. ☐ Ensure environmental sustainability
8. ☐ Develop a global partnership for development

E. More Information :

Please provide a website for this activity

Website (URL) : http://www.ethiopia.child_mortality

F. Geographical Coverage* :

Please tick a box to indicate the geographical coverage

○ Local ○ National ○ Regional ◉ International

Please specify coverage : Ethiopia.Eritrea

G. Timescale * :

Please tick a box to indicate the timescale of the activity

○ Completed ○ Planned for future ○ Ongoing

Specify dates using the format day/month/year (dd/mm/yyyy) :

From: 01/05/2010 To: 30/04/2013

H. Activity Type * :

Please tick one or more boxes to indicate the type of activity described above

☐ Project ☐ Programme ☐ WSIS Thematic Meeting ☐ Conference ☐ Publication ☐ Training initiative

☐ Guidelines ☐ Tool-kit ☐ Website ☐ Database

Other (please specify) :

Problems with online questionnaires

- Sampling is problematic if population size is unknown
- Preventing individuals from responding more than once can be a problem
- Individuals have also been known to change questions in email questionnaires

Deploying online questionnaires

- Plan the timeline
- Design offline
- Program/complete online template
- Test the survey to make sure that it behaves as you would expect
- Test it with a group that will not be part of the survey to check that the questions are clear
- Recruit participants

Observation

Observation (1)

- Direct observation in the field
  - Structuring frameworks
  - Degree of participation (insider or outsider)
  - Ethnography https://www.qualtrics.com/experience-management/research/ethnographic-research/
- Direct observation in controlled environments
- Indirect observation: tracking users' activities
  - Diaries
  - Interaction logging
  - Web analytics
  - Video and photographs collected remotely by drones or other equipment

Observation (2)

Figure 8.8 Mars Exploration Rover
Source: Reproduced by permission of NASA Jet Propulsion Laboratory (NASA-JPL).

Structuring frameworks to guide observation

- Three easy-to-remember parts:
  - **The person:** Who?
  - **The place:** Where?
  - **The thing:** What?

Structuring frameworks to guide observation

- A more detailed framework (Robson, 2014):

  Space: What is the physical space like and how is it laid out?

  Actors: What are the names and relevant details of the people involved?

  Activities:

  What are the actors doing and why?

  Objects: What physical objects are present, such as furniture

  Events:

  Is what you observe part of a special event?

  Time: What is the sequence of events?

  Goals: What are the actors trying to accomplish?

  Feelings:

  What is the mood of the group and of individuals?

Planning and conducting observation in the field

- Decide on how involved you will be: from **passive observer to active participant**
- How to gain acceptance
- How to handle sensitive topics, for example, culture, private spaces, and so on
- How to collect the data:
  - What data to collect
  - What equipment to use
  - When to stop observing

Ethnography

- Ethnography is a philosophy with a set of techniques that include participant observation and interviews
- Ethnographers immerse themselves in the culture that they study
- A researcher's degree of participation can vary
- Analyzing video and data logs can be time-consuming
- Collections of comments, incidents, and artifacts are made

More on Ethnography

- Co-operation of people being observed is required
- Informants are useful
- Data analysis is continuous
- Interpretivist technique
- Questions get refined as understanding grows
- Reports usually contain examples

https://www.indeed.com/career-advice/career-development/examples-of-ethnography

https://pdxscholar.library.pdx.edu/cgi/viewcontent.cgi?article=5428&context=open_access_etds

Ethnography Examples

- Observing a group of children playing
- Observing employees in a corporate office
- Observing medical personnel in a high-volume hospital
- Observing an indigenous village
- Observing a high school classroom
- Observing motorcycle riders

Online Ethnography

- Virtual, Online, Netnography
- Online and offline activity
- Interaction online differs from face-to-face
- Ethical considerations and presentation of results are different

Online ethnography: Read this (1)

Doing Ethnography Online

As collaboration and social activity online have increased, ethnographers have adapted their approach to study the various forms of computer-mediated activity (Rotman et al., 2013; Bauwens and Genoud, 2014). This practice has various names: online ethnography (Rotman et al., 2012), virtual ethnography (Hine, 2008), netnography (Kozinets, 2020), mobile ethnography (Muskat, 2020) and digital ethnography (Pink et al., 2016). Whether online or offline techniques are used, or a combination of both, depends on the community or activity being studied. Since participant observation is a hallmark of ethnography, it is not surprising that ethnography is increasingly used to understand people's behavior in online social spaces both for its own sake and to inform the design of the technology that supports interaction online.

Why is it necessary to distinguish between online and face-to-face ethnography? It is important because interaction online is different from interaction in person (Winter and Lavis, 2019). For example, communication in person is richer (through gesture, facial expression, tone of voice, and so on) than online communication, and participants may feel that anonymity is more easily achieved when communicating online. In addition, virtual worlds have a persistence, due to regular archiving, that does not typically occur in face-to-face situations. This makes characteristics of the communication different, including how ethnographers behave and how they report their findings. Ethical issues need to be considered for any research involving humans and living organisms, but when working online, it can be easy to forget that people exist behind the textual comments and avatars being analyzed. It is therefore important to “listen” carefully, reflect, and remember ethical practices (Winter and Lavis, 2019).

Online ethnography: Read this (2)

For observational studies in large social spaces, such as digital libraries or Facebook, there are different ethical issues to consider. For example, obtaining informed consent requires different tactics, and the presentation of results needs to be modified too. Instead of relying on individuals explicitly agreeing to take part in a study, the researcher must rely on implicit agreement by their continuing to take part. Quotes from participants in the community, even if anonymized in the report, can easily be attributed by a simple search of the community archive or the IP address of the sender, so care is needed to protect their privacy.

Special tools may be developed to support ethnographic data collection. Mobilab is an online collaborative platform that was developed for citizens living in Switzerland to report and discuss their daily mobility during an eight-week period using their mobile phones, tablets, and computers (Bauwens and Genoud, 2014). Mobilab enabled the researchers to more easily engage in discussion with participants on a variety of topics, including trucks parking on a bikeway.

Observations and materials that might be collected for ethnography (1) (Crabtree, 2003)

- Activity or job descriptions
- Rules and procedures that govern particular activities
- Descriptions of activities observed
- Recordings of the talk taking place between parties
- Informal interviews with participants explaining the detail of observed activities

Observations and materials that might be collected for ethnography (2) (Crabtree, 2003)

- Diagrams of the physical layout, including the position of artifacts

- Other information collected when observing activities:
  - Photographs of artifacts (documents, diagrams, forms, computers, and so forth)
  - Videos of artifacts
  - Descriptions of artifacts
  - Workflow diagrams showing the sequential order of tasks
  - Process maps showing connections between activities

Activity (1)

- **Scenario 1:** A usability consultant joins a group of tourists who have each been given a wearable navigation device that fits onto a wrist strap to test on a visit to Stockholm. After sightseeing for the day, they use the device to find a list of restaurants within 2 kilometers of their current position. Several are listed, and they find the phone numbers of a few, call them to ask about their menus, select one, make a booking, and head off to the restaurant. The usability consultant observes some difficulty operating the devices, especially on the move. Discussion with the group supports the evaluator's impression that there are problems with the interface, but on balance the device is useful, and the group is pleased to get a table at a good restaurant nearby.

- **Scenario 2:** A usability consultant observes how participants perform a preplanned task using the wearable navigation device in a usability laboratory. The task requires the participants to find the phone number of a restaurant called Matisse. It takes them several minutes to do this, and they appear to have problems. The video recording and interaction log suggest that the interface is quirky and the audio interaction is of poor quality. This is supported by participants' answers on a user satisfaction questionnaire.

1. **What are the advantages and disadvantages of these two types of observation?**
2. **When might each type of observation be useful?**

Activity (2): Sample Answers

1. The advantages of **observation in the wild** are that the observer saw how the device could be used in a real situation to solve a real problem. The disadvantage is that the observer was an insider in the group, so how objective could they be?
The **lab study** is easier to replicate, so several users can perform the same task, specific usability problems can be identified, participants' performance can be compared, and averages for such measures as the time it took to do a specific task and the number of errors can be calculated. The observer could also be more objective as an outsider. The disadvantage is that the study is artificial and says nothing about how the device would be used in a natural setting.

2. Both types of observation have merits, depending on the goals of the study. The **lab study** is useful for examining details of the interaction style to make sure that usability problems with the interface are diagnosed and corrected. **The other study** reveals how the navigation device is used in a natural setting and how it integrates with or changes people's behavior. According to Kjeldskov and Skov (2014), there is no definitive answer to which kind of study is preferable for mobile devices. They suggest that the real question is when and how to engage with longitudinal, i.e., long-term, in-the-wild studies.

Activity to try later

ACTIVITY 8.7

1. Find a small group of people who are using any kind of technology, for example, smartphones, household appliances, or computer games, and try to answer the question, “What are these people doing?” Watch for three to five minutes, and write down what you observe. When finished, note how it felt to be doing this and any reactions in the group of people observed.
2. If you were to observe the group again, what would you do differently?
3. Observe this group again for about 10 minutes using the detailed framework given earlier.

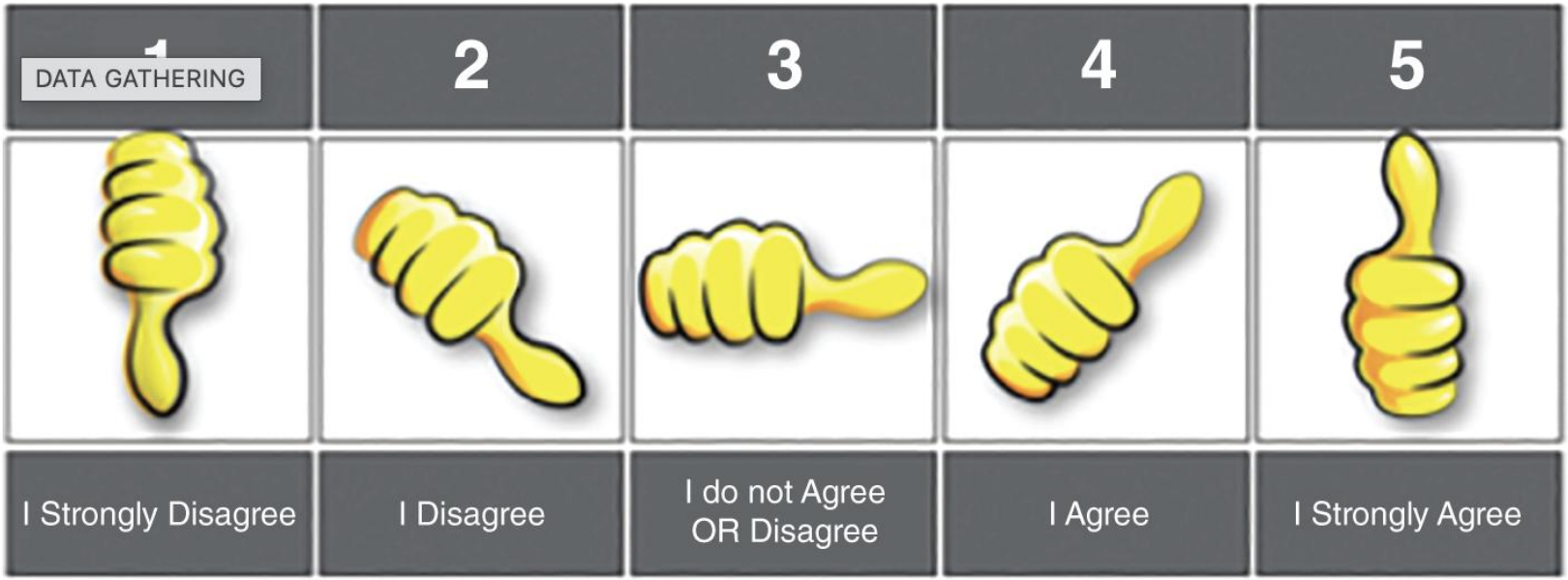
Comment

1. What problems did this exercise highlight? Was it hard to watch everything and remember what happened? How did the people being watched feel? Did they know they were being watched? Perhaps some of them objected and walked away. If you didn't tell them that they were being watched, should you have?
2. The initial goal of the observation, that is, to find out what the people are doing, was vague, and chances are that it was quite a frustrating experience not knowing what was significant and what could be ignored. The questions used to guide observation need to be more focused. For example, you might ask the following: What are the people doing with the technology? Is everyone in the group using it? Are they looking pleased, frustrated, serious, happy? Does the technology appear to be central to their activity?
3. Ideally, you will have felt more confident this second time, partly because it is the second time doing some observation and partly because the framework provided a structure for what to observe. ■

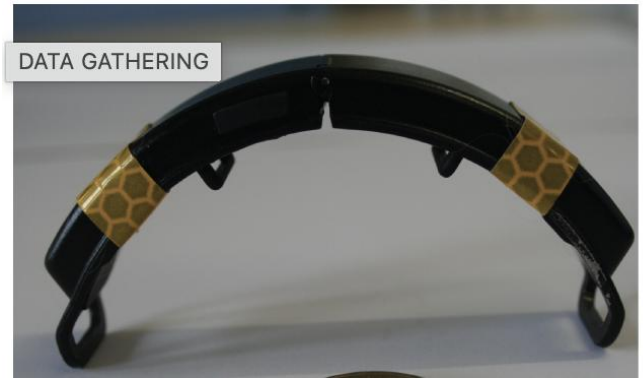
Choosing and combining techniques

Technique	Good For	Kind of Data	Advantages	Disadvantages
Interviews	Exploring issues	Some quantitative but mostly qualitative	Interviewer can guide interviewee if necessary. Encourages contact between researchers and participants.	If interview takes place away from participants' own environment, it may seem artificial and they may be intimidated.
Focus groups	Collecting multiple viewpoints	Some quantitative but mostly qualitative	Highlights areas of consensus and conflict. Encourages contact between researchers and participants.	Possibility of dominant characters and "groupthink."
Questionnaires	Answering specific questions	Quantitative and qualitative	Can reach a wide range of people with low resource requirements.	Response rates may be low. Unless carefully designed, the responses may not provide suitable data.
Direct observation in the wild	Understanding context of activity	Mostly qualitative	Observing gives insights that other techniques don't provide.	Very time-consuming. Huge amounts of data can be produced.
Direct observation in a controlled environment	Capturing the detail of what individuals do	Quantitative and qualitative	Can focus on the details of a task without interruption.	Results may have limited use in the natural setting because the conditions were artificial.
Indirect observation	Observing activity without researcher being present; data may be captured automatically or by participant.	Quantitative (logging) and qualitative (diary)	Participant isn't distracted by the data gatherer; automatic recording means that it can extend over long periods of time.	A large amount of quantitative data needs tool support to analyze (logging); participants' memories may exaggerate (diary).

Adapting techniques for different settings (1)



Adapting techniques for different settings (2)



Summary

- Data gathering sessions should have clear goals
- An informed consent may be needed
- Five key issues of data gathering are: goals, choosing participants, triangulation, participant relationship, pilot
- Data may be recorded using handwritten notes, audio or video recording, a camera, or any combination of these
- Interviews may be structured, semi-structured, or unstructured
- Focus groups are group interviews
- Questionnaires may be on paper, online, or telephone
- Observation may be direct or indirect, in the field, or in controlled settings
- Techniques can be combined depending on the study focus, participants, nature of technique, and available resources and time